TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Fibre Strength by Pressley Tester, Count of Yarn and Twist of Single and Double Yarn

Three bench tests that link fibre to yarn: how strong the fibre bundle is, how thick the yarn spun from it is, and how much twist holds that yarn together.

TE-3104 · QUALITY EVALUATION OF FIBRE AND YARN SESSIONAL · EXPERIMENT 2 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to determine (a) the bundle strength of cotton fibres with the Pressley tester, expressed as the Pressley index, (b) the count (linear density) of a yarn from a full lea weighed on the wrap reel and balance, and (c) the twist in turns per inch of a single yarn and of a doubled (plied) yarn by the untwist method, including the twist multiplier of the single yarn.

1.1 Pressley bundle strength

Single cotton fibres are too fine and variable to break one by one in a routine mill laboratory, so strength is measured on a flat, parallel bundle. In the Pressley tester a small ribbon of fibres is clamped between two jaws whose faces meet with **zero gauge length** — the fibres are gripped at exactly the break zone, so virtually every fibre in the bundle is loaded and broken. A weighted beam rolls along a graduated incline until the bundle ruptures; the beam position reads the breaking load. Dividing by the bundle mass removes the effect of bundle size:

$$\text{Pressley index} = \text{breaking load (lb)} / \text{bundle mass (mg)}$$

The index is conventionally reported on a scale equivalent to thousands of pounds per square inch (thousand psi) of bundle cross-section. As a guide: below 70 is weak, 70–80 fair, 80–90 strong, above 90 very strong for upland cotton. An approximate conversion to tenacity is sometimes quoted:

$$\text{Tenacity (g/tex)} \approx \text{Pressley index} / 3.19 \quad \text{(approximate only)}$$

It must be labelled approximate: zero-gauge bundle strength and gauge-length fibre tenacity are related but not identical properties, because longer gauge lengths admit weak places that the zero-gauge clamp suppresses.

1.2 Yarn count

Count expresses yarn fineness. Two families of systems exist. In **direct** systems (tex, denier) the number is the mass of a fixed length — bigger number, coarser yarn. In **indirect** systems (English cotton count Ne, metric count) the number is the length per unit mass — bigger number, finer yarn. The English cotton count is the number of 840-yard hanks (leas) that weigh one pound:

$$\text{Ne} = \text{length (yd)} / 840 \div \text{mass (lb)} = (L / 840) / (W / 453.6)$$

with W in grams. The standard test length is one lea of 120 yd wound on a wrap reel of 1.5 yd circumference (80 wraps). Count is always reported at the standard regain: a count measured on yarn conditioned in a damp room is a count of yarn plus weather, so conditioning before weighing is part of the test, not a refinement of it. The tex equivalent of a cotton count is tex = 590.5 / Ne.

1.3 Twist in single and double yarn

Twist binds the fibres of a spun yarn into a coherent, strong strand. It is measured in turns per inch (tpi). The working quantity in spinning is the **twist multiplier** (TM, also called twist factor in the cotton system), which states the twist appropriate to a given count:

$$\text{tpi} = \text{TM} \times \sqrt{\text{Ne}} \quad \rightarrow \quad \text{TM} = \text{tpi} / \sqrt{\text{Ne}}$$

Typical cotton multipliers are about 3.5–4.0 for soft knitting yarns, 4.5–5.0 for medium weaving yarns and above 5 for hard-twisted warp and crepe yarns. In the **untwist method** a known gauge length (10 in here) is clamped in a twist tester, one clamp rotates, and the operator untwists until the fibres lie parallel — judged by slipping a needle through the strand — and reads the turns removed. For a doubled (two-ply) yarn the same instrument untwists the ply twist; ply twist is normally inserted opposite in direction to the singles twist and at a lower level, chosen to balance the yarn so the fabric does not skew and the plied yarn hangs without snarling.

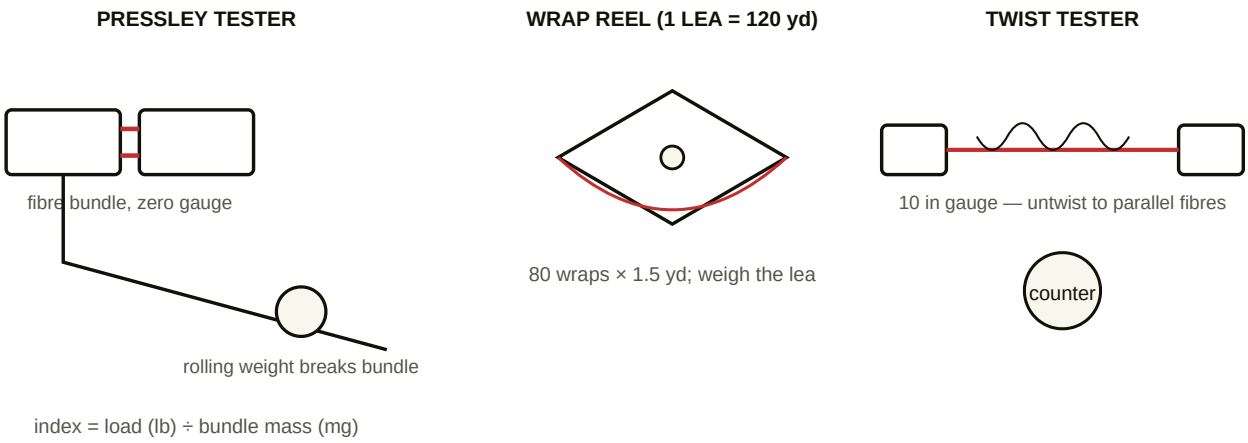


Figure 1 — The three instruments of this sheet: Pressley tester with zero-gauge bundle clamps and rolling beam (left), wrap reel winding a 120-yard lea for count (centre), and twist tester with fixed and rotating clamps over a 10-inch gauge (right).

2. Apparatus and procedure

- **Apparatus:** Pressley tester with clamps, vice, razor blade and torsion balance (0.01 mg or the standard bundle balance); wrap reel (1.5 yd circumference) with counter, yarn package or cop, analytical balance; twist tester with 10 in gauge, revolution counter, needle and magnifier; conditioning facilities.
- **Pressley:** comb a small tuft parallel, clamp a ribbon in the jaws, trim the protruding fibres flush with a razor, and release the rolling beam so it loads the bundle to rupture. Read the breaking load, remove and weigh the broken bundle, and compute the index. Test at least five bundles from different tufts and average; discard any break that slips from a jaw rather than rupturing mid-bundle.
- **Count:** condition the yarn, then wind one lea (80 wraps = 120 yd) on the wrap reel at uniform, light tension — firm enough to lie on the reel, never stretched. Tie off the lea, remove and weigh it. Wind and weigh at least three leas from different parts of the package and average the count. If the yarn is tested at other than standard regain, correct the mass to 8.5% regain before calculating.
- **Twist, single yarn:** mount 10 in of yarn between the clamps under a light standard tension, note the direction of twist (S or Z — cotton singles are almost always Z). Rotate the moving clamp to untwist, testing with the needle until the fibres are parallel along the whole gauge; read the turns. Repeat at least ten times at spaced positions along the yarn and average.
- **Twist, double yarn:** mount the plied yarn the same way and untwist the ply structure until the two singles lie parallel and separate cleanly with the needle; read the ply turns. Note that the direction is normally opposite to the singles twist (S ply over Z singles is the usual balanced construction).

3. Worked calculation

Pressley strength. Five bundles average a breaking load and mass giving a Pressley index of approximately **84** (thousand psi).

$$\text{Pressley index} \approx 84 \rightarrow \text{classification: strong cotton}$$

$$\text{Approximate tenacity} \approx 84 / 3.19 \approx 26 \text{ g/tex (approximate conversion)}$$

An index of 84 places the cotton in the strong band — consistent with a yarn count in the medium range spun at good efficiency, and with the single-yarn tenacity found in Experiment 5 of this course once spinning losses are allowed for.

Count. One lea of 120 yd weighs 2.16 g = 2.16/453.6 lb.

$$Ne = (120 / 840) \div (2.16 / 453.6) = 0.14286 \div 0.004762 = 30.0$$

The yarn is exactly **30 Ne** — the nominal count used throughout the remaining calculations of this sheet and of Experiment 5 (tex = 590.5/30 ≈ 19.68 tex).

Twist, single yarn. Untwist test, 10 in gauge: 285 turns counted.

$$\text{Twist} = 285 / 10 = 28.5 \text{ tpi}$$

$$\text{Twist multiplier} = 28.5 / \sqrt{30} = 28.5 / 5.477 \approx 5.20$$

A multiplier of 5.20 is a hard twist — this is a warp-type yarn, twisted for strength and abrasion resistance rather than softness. Expect a lively yarn that would snarl if cut and released, and a firm fabric hand.

Twist, double yarn. Ply untwist, 10 in gauge: 182 turns counted.

$$\text{Ply twist} = 182 / 10 = 18.2 \text{ tpi (direction opposite to the singles)}$$

Note the design logic: the ply twist (18.2 tpi) is well below the singles twist (28.5 tpi) and opposite in direction, so it partly untwists the singles as it is inserted. The residual torque of the two components roughly cancels — the plied yarn is balanced, hangs in an open loop without kinking, and will produce a fabric that does not skew after washing.

> **CROSS-CHECK**
>
> Count links all three tests: √Ne converts measured tpi into the multiplier that spinners actually set on the machine, and the same 30 Ne gives tex = 19.68 for the tenacity calculation in Experiment 5. An error in the lea weighing therefore propagates into every later result — which is why count is weighed on three leas, conditioned, and averaged.

4. Precautions

- Pressley bundles must be parallel, of even density, and trimmed truly flush; a tapered bundle concentrates stress and breaks early, and a bundle that slips in the jaws gives a load that measures friction, not strength.
- The wrap reel tension must be light and steady. Yarn is elastic: a hard-pulled lea is a stretched lea, longer than 120 yd of yarn at rest, and the count comes out falsely fine.
- Condition before weighing — at 65% RH a cotton lea holds about 8% water; an unconditioned lea weighed on a dry day can shift the count by a full count number.
- In the untwist test judge the end point by the needle test along the whole gauge, not by eye alone, and take the reading at the same tension each time; over-tensioning stretches the yarn and adds apparent turns, under-tensioning lets it sag and corkscrew.
- Take twist readings at spaced intervals — twist runs preferentially into the thin places of a yarn, so ten readings clustered in one metre describe that metre, not the package.
- Report the direction of twist (S or Z) with every result; a bare tpi figure without direction is half a result, and for plied yarn it conceals whether the construction is balanced at all.

Practice (viva preparation)

1. Why does the Pressley tester use zero gauge length, and how does that affect the strength value compared with a gauge-length test?
2. A bundle breaks at a load corresponding to an index of 84. Classify the cotton and give its approximate tenacity.
3. A lea of 120 yd weighs 1.80 g. Find the count in Ne.
4. A 20 Ne yarn is to be spun with a twist multiplier of 4.8. What twist in tpi must the machine insert?
5. Why is the ply twist of a doubled yarn inserted in the opposite direction to, and at a lower level than, the singles twist?
6. Why is count described as an indirect system in Ne but a direct system in tex?

Answer key

1. At zero gauge length the clamps meet at the break zone, so every fibre is gripped and loaded almost equally and the test measures the fibre material itself rather than the weakest flaw in a length of bundle. Values are therefore higher and less variable than gauge-length results, which find the weak places — one reason the Pressley-to-tenacity conversion is only approximate.
2. 84 thousand psi is in the strong band (80–90). Approximate tenacity = 84/3.19 ≈ 26 g/tex, labelled approximate because of the zero-gauge effect described above.

3. Ne = (120/840) ÷ (1.80/453.6) = 0.14286 ÷ 0.003968 = 36.0 Ne. (Equivalently, count is inversely proportional to lea mass: 30 Ne at 2.16 g → 2.16/1.80 × 30 = 36.)
4. tpi = TM × √Ne = 4.8 × √20 = 4.8 × 4.472 ≈ 21.5 tpi.
5. Opposite-direction ply twist unwinds part of the singles twist as it is applied, so the torques of the components oppose and largely cancel: the doubled yarn is balanced, does not snarl, and the fabric does not skew. A lower level is used because fully cancelling 28.5 tpi of singles twist with 28.5 tpi of ply twist would over-soften and weaken the structure; about two-thirds (here 18.2 tpi) is the practical compromise — verified in this experiment's data.
6. In Ne the number counts lengths per unit mass, so a finer yarn (more length in a pound) carries a bigger number — indirect. In tex the number is the mass of a fixed 1,000 m length, so a coarser yarn carries the bigger number — direct. tex = 590.5/Ne converts between them for cotton counts.

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